

Team Roles & Responsibilities

AI-Augmented Startup Pitch Analyzer

Team Members

Name	Role
Kushagra	Full-Stack Developer
Kritika	Frontend Developer + Documentation Lead

Kushagra — Backend & AI Development

Backend Development (Flask + Python)

Task	Details
Flask App Setup	Configured app.py — Flask app, blueprints, CORS, JWT, SQLAlchemy
Extensions	Created extensions.py — shared db, jwt, cors instances
Database Models	Built Startup, Analysis, and User SQLAlchemy models with all fields
Auth Routes	Built /api/auth/register and /api/auth/login with bcrypt password hashing
Startup Routes	Built full CRUD for /api/startups — handles multipart form data, PDF and logo file uploads
Analysis Routes	Built /api/generate-analysis and /api/analysis/:id/report endpoints
Health Route	Built /api/health status check endpoint
Controllers	Implemented business logic layer separated from route handlers
Middleware	Built JWT authentication middleware to guard all protected API routes
DB Migration	Implemented ensure_database_columns() for safe column additions in SQLite and PostgreSQL
File Storage	Managed upload folder for storing pitch deck PDFs and logos

AI Integration (OpenAI + PyMuPDF)

Task	Details
PDF Text Extraction	Integrated PyMuPDF (fitz) to read and extract all text from uploaded pitch decks
AI Call #1 — VC Analysis	Built the GPT-4o-mini prompt for VC scoring: 8 scores + strengths, weaknesses, risks, opportunities, investment recommendation, due diligence questions
AI Call #2 — Market Intelligence	Built the GPT-4o-mini prompt for market research: competitors, SWOT, TAM/SAM/SOM, industry trends, strategic recommendations
JSON Parsing	Built response normalization functions to cleanly parse AI JSON output

Task	Details
Fallback System	Built mock/fallback data that returns when OpenAI API key is not configured
PDF Report Generation	Implemented <code>generate_pdf_report()</code> using <code>fpdf</code> — auto-generates downloadable evaluation PDF

Deployment & Configuration

Task	Details
Production Server	Configured Gunicorn as the WSGI production server
Database	Set up PostgreSQL for production and SQLite for local development
Environment Variables	Created <code>.env.example</code> template for all required configurations
Runtime	Created <code>runtime.txt</code> for Python version pinning on deployment

Kritika — Frontend Development + Documentation Lead

Frontend Development (React + Vite)

Task	Details
Project Setup	Initialized React + Vite project, configured Tailwind CSS and PostCSS
App Routing	Set up React Router v6 with all routes and role-based protected routing
Auth Context	Built AuthContext.jsx — global login/logout state using React Context API
Protected Routes	Built ProtectedRoute.jsx — blocks page access based on JWT and user role
Navbar	Designed and built Navbar.jsx with auth-aware navigation links
Axios Setup	Configured axios.js with base URL and automatic JWT Authorization headers
Home Page	Built HomePage.jsx — the landing and welcome screen
Login Page	Built LoginPage.jsx — login form with JWT handling
Register Page	Built RegisterPage.jsx — registration form with role selection (Founder / Investor)
Profile Page	Built ProfilePage.jsx — user profile and account settings
Founder Page	Built FounderPage.jsx — Founder role landing screen
Founder Dashboard	Built FounderDashboardPage.jsx — pitch submission form (13 fields + file uploads), KPI snapshot, Radar chart, analysis result cards
Investor Dashboard	Built InvestorPage.jsx — analytics dashboard with 4 charts, 4 KPI cards, startup pipeline, sidebar, investor actions
Chart Integration	Integrated Chart.js via react-chartjs-2: Radar, Bar (horizontal + vertical), Line, Doughnut
Search & Filter	Implemented search bar, category filter, sort dropdown, and pagination on both dashboards
PDF Download	Implemented frontend download logic for AI-generated PDF reports

Documentation (Full Ownership)

Task	Details
README File	Wrote the complete project README — setup guide, features, tech stack, flowcharts, deployment link
Project Report	Wrote the full internship project report covering objective, scope, methodology, and results
Problem Statement	Defined and documented the core problem the project addresses
Solution Statement	Documented how the platform solves the problem for Founders and Investors
System Flowchart	Documented the complete system flow — user visit → form → AI → dashboard

Task	Details
Data Flow Diagram	Documented the full data flow from browser through Flask to OpenAI and back
API Documentation	Documented every API endpoint — method, route, request body, response, auth
Database Schema	Documented the users, startups, and analyses tables with all columns
AI Pipeline Docs	Documented the 5-step AI pipeline triggered on form submit
AI Output Docs	Documented all 15+ AI outputs: 8 scores, 7 qualitative fields, 11 market intelligence fields
Folder Structure	Documented every folder and file in the project with its purpose
Feature Docs	Documented Founder journey, Investor dashboard features, form fields, auth flow
Technology Research	Researched and justified the tech stack — React, Flask, GPT-4o-mini, PyMuPDF, Chart.js
Competitor Research	Researched existing pitch evaluation tools and identified gaps this project fills
Presentation Slides	Prepared the slide deck for the internship presentation
Demo Script	Wrote the walkthrough script for the live app demo
Roles Document	Created this team roles and responsibilities file
Final Report	Compiled and formatted the final internship submission report

Work Division Summary

Kushagra	Kritika
■ Flask Backend (all routes)	■ React Frontend (all pages)
■ Database Models (SQLAlchemy)	■ App Routing + Protected Routes
■ JWT Authentication (bcrypt)	■ Auth Context + Navbar
■ REST API Endpoints (CRUD)	■ Founder Dashboard (Form + Charts)
■ File Upload Handling	■ Investor Dashboard (Charts + Pipeline)
■ AI Integration (GPT-4o-mini)	■ Chart.js Visualizations
■ PDF Text Extraction (PyMuPDF)	■ Search, Filter, Sort, Pagination
■ PDF Report Generation (fpdf)	■ PDF Download (Frontend)
■ Fallback / Mock Data System	■ README File
■ DB Migration Logic	■ Project Report
■ Gunicorn Production Setup	■ System Flowchart & Data Flow Diagram
■ PostgreSQL Configuration	■ API Documentation
■ Environment Variable Config	■ Database Schema Documentation
	■ AI Pipeline Documentation
	■ Technology & Competitor Research
	■ Presentation Slides + Demo Script
	■ Final Internship Report

Shared Responsibilities

Responsibility	Kushagra	Kritika
Project planning & feature scoping	■	■
Feature brainstorming	■	■
Testing the live application	■	■
Final presentation delivery	■	■
Bug fixing and QA	■	■

Technologies Used

Kushagra

Flask · Flask-SQLAlchemy · Flask-JWT-Extended · Flask-CORS · bcrypt · OpenAI API (GPT-4o-mini) · PyMuPDF · fpdf · Unicorn · PostgreSQL · SQLite · python-dotenv

Kritika

React 18 · Vite · React Router v6 · Tailwind CSS · Chart.js · react-chartjs-2 · Axios · Markdown · Microsoft Word / Google Docs · PowerPoint / Google Slides

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Team: Kushagra (Backend + AI) & Kritika (Frontend + Documentation)